

CHE 313: Transport Phenomena III

Instructor: Gang Cheng (gancheng@uic.edu), Chemical Engineering

Fall 2019: Monday, Wednesday & Friday, 1:30-2:20 pm, EIB 124

Course Aims

Introduction to the fundamentals of separation processes and their applications in a variety of technologies (i.e., distillations, absorption, extraction, membranes). Includes methods for the analysis of separation technologies.

Prerequisite: CHE 301.

Course Learning Outcomes

By the end of this course, students will be able to:

- Be familiar with separation principles, design, separation mechanisms, processes, and applications
- Understand the underlying thermodynamic and mass transfer principles
- Explore the use of various types of separation techniques
- Explore the use of alternative separation techniques to the existing ones

ABET Student Learning Outcomes covered in this course:

- (1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- (2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- (3) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Textbook and General References

Textbook

- Christie John Geankoplis, Allen Hersel, Daniel H. Lepek, *Transport processes and Separation Process Principles*, 5th edition, Prentice Hall, 2018

References

- Seader, J.D et al., *Separation Processes Principles*, 3rd edition, John Wiley & Sons, New Jersey, 2011

Office Hours:

TA: Seyyedfaridoddin Fattahpour, sfatta2@uic.edu

TA office hours: Opon office hours

Professor office hours: Monday, Wednesday & Friday, 12:30-1:20 pm, EIB 254

Course Website:

The course syllabus, homework assignments, and all handouts will be posted on the blackboard. Students will be responsible for keeping up to date with the information on this website

Grading Procedure:

Homework	10%
Quizzes	15%
Exam1	20%
Exam2	20%
Final Exam	35%

Homework:

Homework will be assigned every week along with the due date. You are encouraged to work through all the problems so that you can understand the materials and prepare themselves for the quizzes and exams. Final answers and compact solutions will be posted on the blackboard. But detailed written solutions will not be provided. You may consult anyone when working these problems. However, direct copying of another student's work will be considered an honor code violation. Without medical excuse, late homework will not be accepted.

Quizzes:

Quizzes will be occasionally given to encourage progressive mastery of the course content. There will be a total five in-class quizzes. A student who is absent (regardless of the reason) will not be given a make-up quiz. All quizzes are open-book and open-notes.

Exams:

Exams and quizzes will cover all material up to and including the last lecture before the test unless otherwise specified. Each test will emphasize the material since the last test. There will be no make-up exams. All exams are closed-book and closed-notes

Course Outline

- Introduction
 - Course outline and assessment
 - Introduction to separation processes
- Flash Distillation
- Binary distillation
- Multicomponent distillation
- Batch distillation
- Stage and packed column design
- Absorption and stripping
- Liquid-liquid extraction
- Membrane separations